Residue intersections in the supplied Higman–Sims geometries

Private bounded computational proof

1 The roster and the retained classification

This note processes the exact supplied Higman–Sims permutation families in the supplementary archive to L096. The four files contain respectively $44, 67, 12, 1$ active entries of ranks $3, 4, 5, 6$. The parser preserves each literal final reordering and records its construction index separately. Every one of these 124 entries is checked for flag transitivity, firmness, residual connectivity, residual weak primitivity, local 2-transitivity and the rank-two intersection property. No prior Higman–Sims census is used to assume any of these predicates.

Entire-family equivalence under group automorphisms and type permutations reduces this supplied roster to $30, 41, 8, 1$ distinct families. There are 44 classes represented twice and 36 represented once. These are exhaustive conclusions about the supplied roster. The assertion that it exhausts the intended classified geometries is retained from the earlier classification cited by L096, reference [21]: D. Leemans, *Supplement to "The residually weakly primitive geometries of HS"* (2003). We do not replace that classification by a count match. In particular, no published historical suffix such as 4.1 is identified with an active archive position: the archive itself gives examples where they differ.

2 The common finite criteria

The accepted orbit–double-coset shadow theorem and the accepted sparse right-coset adapter are used with their full hypotheses. For a family $(G_i)_{i\in I}$ in G, put $G_J = \bigcap_{j\in J} G_j$ and $G_\varnothing = G$. The base residue at a flag of types J has ambient G_J and parabolics $G_{J\cup\{i\}}$, $i \notin J$. The separately checked flag transitivity transports this residue to every flag of those types. All proper subsets J are processed; there are $[G : G_J]$ translates. Rank-one residues supply the panel tests; only ranks 2 through $|I|$ enter the intersection and weak-intersection quantifiers. Rank zero is not subjected to an existential weak-intersection test.

For each residue and each type, the accepted algorithm tests every required orbit representative, retaining actual nonsingleton shadow intersection witnesses when the test fails. Sparse shadows are sets of canonical right-coset representatives. The empty flag's whole universe is used by its exact cardinality, avoiding unnecessary materialization; this is the previously proved adapter, unchanged byte for byte here. The two new implementation interfaces are given with proofs below.

2.1 Primitivity without constructing a large coset action

Let $H \le G$ be finite. The transitive action of G on its right cosets of H is primitive of degree at least two if and only if $H < G$ and

$$\langle H, x\rangle = G \quad \text{for every representative } x \notin H \text{ of an } (H,H)\text{-double coset in } G. \tag{*}$$

This is the standard block–overgroup correspondence together with an explicit double-coset test. Here is the complete argument. A block containing the base coset H is the orbit of H under a subgroup K with $H \leq K \leq G$: its setwise stabilizer is K, and transitivity shows the block is exactly $\{Hk : k \in K\}$. Conversely this set is a block for each such K. Hence degree-at-least-two primitivity is exactly maximal properness of H. If $g = h_1 x h_2$, with $h_1, h_2 \in H$, then $\langle H, g\rangle = \langle H, x\rangle$. A proper intermediate subgroup therefore contains an element outside H and supplies a representative violating (*); conversely any violating representative gives the proper intermediate subgroup $\langle H, x\rangle$. The argument does not require a faithful coset action or a core-free stabilizer. The case $H = G$ must be rejected separately: its degree-one action is outside the convention and would otherwise satisfy (*) vacuously.

The finite implementation enumerates the complete double-coset roster, forms each indicated generated subgroup and compares its order with $|G|$. Every negative nontrivial case records the outside element and actual intermediate subgroup generators. For a residue it tests types in increasing coset degree until a primitive type is found, or exhausts all types. This preserves the existential weak-primitivity quantifier. Applying it to all base residues is legitimate by the separately checked flag transitivity; the residue common theorem supplies their actual stabilizers and translates. No previously classified geometry predicate is used by this eligibility test.

2.2 Exact family transport with necessary order filters

Let $A \leq \mathrm{Aut}(G)$ be a finite, explicitly computed transformation group, and let $\mathcal{H} = (H_1, \ldots, H_n)$ and $\mathcal{K} = (K_1, \ldots, K_n)$ be ordered families. We seek an element of A carrying the first family to a permutation of the second. Put $H_J = \bigcap_{j \in J} H_j$, with $H_\varnothing = G$, and define K_J similarly. If a proposed type permutation sends H_i to $K_{\pi(i)}$, then necessarily

$$|H_J| = |K_{\pi(J)}| \qquad (J \subseteq \{1, \ldots, n\}). \tag{**}$$

Indeed an automorphism commutes with intersection and preserves order. Thus a failed equality rejects that type permutation, and a failed comparison of the two multisets of intersection orders rejects every permutation. Satisfying (**) never certifies equivalence.

Fix a surviving target tuple $\mathcal{T}$ and any order $i_1, \ldots, i_n$ in which to place its entries. Start with $C = A$ and $\mathcal{H}' = \mathcal{H}$. At step k find an actual transporter $x \in C$ with $x(H'_{i_k}) = T_{i_k}$. If none exists, reject this target. Otherwise replace every H'_i by $x(H'_i)$ and replace C by the setwise stabilizer of T_{i_k} in C. After the last step no further stabilizer is needed. Success supplies a product of the computed automorphisms taking the entire family to the target.

For completeness, after the first $k-1$ placements all possible further solutions differ from the current partial transporter by an element of the pointwise stabilizer of the already placed *subgroups*. If x is one solution to the next placement, all such solutions differ from x by an element fixing its target subgroup. Hence passing to the stated stabilizer loses no solution. Induction proves both soundness and completeness, for every placement order. The stabilizer after k placements depends only on A and the first k target subgroups, not on the input family or the transporters chosen. It can therefore be cached for repeated comparisons against the same target and placement order. In this implementation larger target subgroups are placed first, with indices breaking ties. This changes computation time, not the criterion. Literal equality of complete tuples is an immediate successful case.

The code loops over *all* type permutations. GAP computes $A = \mathrm{Aut}(\mathrm{HS})$, of order 88 704 000, from the literal permutation group. It uses exact group equality, automorphism images, transporter searches and stabilizers; it does not infer a positive match from orders or a graph label. The first 31 completed roster decisions from the original exact matcher are preserved. The optimized

matcher agrees on 30 nonreflexive positive and negative comparisons drawn from that prefix, and completes the remaining 93 entries. A separate full optimized replay is supplied. GAP's group and automorphism algorithms remain trusted finite computational inputs.

3 Complete residue results

For rank $k \geq 2$, IP_k means every rank-k residue satisfies every directional test, while WIP_k means that in each such residue at least one direction succeeds. Full IP and WIP quantify these conditions over all ranks 2 through the rank of the geometry. The existential direction may vary from residue to residue. In the table, (a, b, c) records IP, WIP but not IP, and failure of WIP.

Rank	Classes	IP_3	IP_4	IP_5	IP_6	Full
3	30	$(8, 5, 17)$	—	—	—	$(8, 5, 17)$
4	41	$(8, 1, 32)$	$(5, 0, 36)$	—	—	$(5, 0, 36)$
5	8	$(1, 0, 7)$	$(1, 0, 7)$	$(0, 1, 7)$	—	$(0, 1, 7)$
6	1	$(0, 0, 1)$	$(0, 0, 1)$	$(0, 0, 1)$	$(0, 0, 1)$	$(0, 0, 1)$

Here each heading IP_k denotes the complete three-way $\mathrm{IP}_k/\mathrm{WIP}_k$ classification just defined; every family also satisfies IP_2. The results agree with all thirteen HS count rows in L096 Tables 1, 3, 5 and 7 (physical pages 5, 7 and 8), and with its rank-six non-WIP statement on physical page 6. Tables 2, 4, 6 and 8 contain historical labels whose exact correspondence remains a separate obligation. The local class identifiers in the machine-readable results are specified by actual archive representatives, not those labels.

Proposition 1. *On the fixed supplied roster, with the retained finite group algorithms, the preceding table and every recorded directional result are complete. The full run covers* 1 136 *base residues including panels, of which* 836 *have rank at least two, and* 2 204 *directional decisions.*

Proof. The parser supplies every literal active tuple. The actual eligibility run covers all 124 tuples and all 1 748 of their proper fixed-type subsets. It tests flag extensions and panel actions, connectivity, rank-two intersections and the genuine primitive predicates established above. Thus all hypotheses required by the residue theorem are verified. The exact family transporter partitions the roster, preserving all incidence and residue predicates under its isomorphisms. Applying the unchanged shadow algorithm to each of its 80 representatives therefore suffices. For every representative the checker reconstructs the complete set of proper subsets, their residue ranks, flag-orbit multiplicities, and every all/exists quantifier from the recorded directional values. Summing the resulting three-way classifications gives the displayed table. No conclusion is drawn from agreement with the printed counts; that agreement is checked afterwards. □

4 Reproducibility and scope

The first six eligibility entries use explicit coset actions. Their completed outputs are preserved from a bounded run that timed out on the next entry. The remaining 118 use the exact maximality criterion; 31 small direct-action comparisons include degree-one exclusions and nonfaithful primitive actions. Likewise the first 31 family decisions are preserved from the initial bounded matcher. Failed continuations and unsuccessful accelerations are retained as execution history, not as mathematical evidence. The successful continuation takes approximately 54 seconds, and the complete shadow

run approximately 21 seconds in the recorded environment; these are replay observations, not asymptotic claims or a comparison to historical timings.

The required whole programs, generated input data, complete successful outputs and validation roles are inventoried separately. The default checker verifies their bindings and reconstructs all quantifiers; an optional bounded replay recomputes eligibility, the complete optimized partition and every shadow decision. The scientific code has no Lean formalization. This bounded note does not award a whole-original L096 recovery, a historical-label identification, a new classification or a compression factor. Standard block and transporter arguments are attributed as such; the separately accepted residue and sparse-shadow proofs remain explicit dependencies.